Setting out to see how to incorporate preference heterogeneity over the product's overall bundle into model via augmenting the U_{ijt} term it is now defined as follows:

$$U_i jt = \sum_j \beta_{ji} X_{jt} + \gamma_i \log(1 + \Xi_{ijt}^2) + \varepsilon_{ijt}$$

$$\text{s.t.}$$

$$\Xi_{ijt} = \sum_j \sqrt{(X_{jt} - \bar{X}_{jt})^2}$$

$$\varepsilon_{ijt} \sim Gumbel(0, 1)$$

This allows for β to vary with the specific characteristics the product holds rather than just a beta for the overall bundle. Variety now takes the euclidean distance of each characteristic of the product's bundle from the amount the consumer has already consumed in the past, allowing for a more nuanced measure of variety that captures the idea that consumers may want to try products that have some different characteristics from their past consumption, but not necessarily products that are completely different from what they have already consumed.

Product	Characteristic 1	Characteristic 2	Total Value
1	6.88	8.16	15.04
2	2.93	0.41	3.34
3	8.12	1.72	9.84
4	9.87	7.20	17.08
5	2.24	2.56	4.81

Table 1: Product characteristics

Results reported:

- Utility graphs
- Regression results on log difference in shares
- Inclusive value tables

The regression results make it seem as if γ is not meaningfully accounting for anything missing from the naive baseline regression. The coefficients on the characteristics are not very different across the baseline and LOV regressions, and the coefficient on the variety term is still pretty far from the true value. I wonder if this is a data issue, where maybe I need more variation in Ξ to be able to identify the γ coefficient. I will try to generate some data with more variation in Ξ by upping to 5 consumers, and see if that helps.

1 Utility Graphs

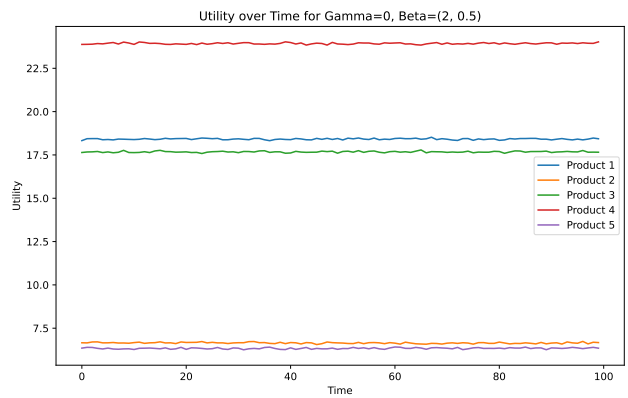

Figure 1: Utility graph for $\gamma = 0$, $\beta_1 = 2$, $\beta_2 = 0.5$

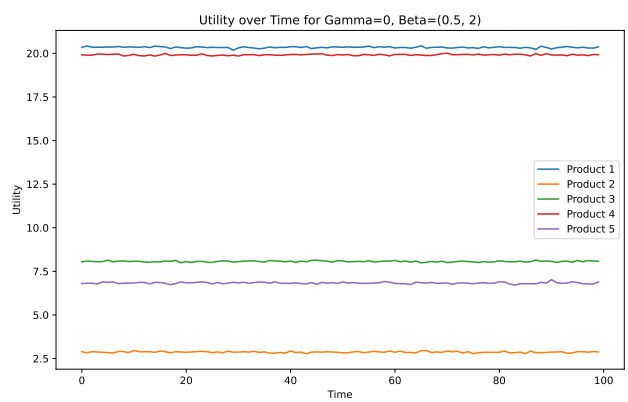

Figure 2: Utility graph for $\gamma = 0$, $\beta_1 = 0.5$, $\beta_2 = 2$

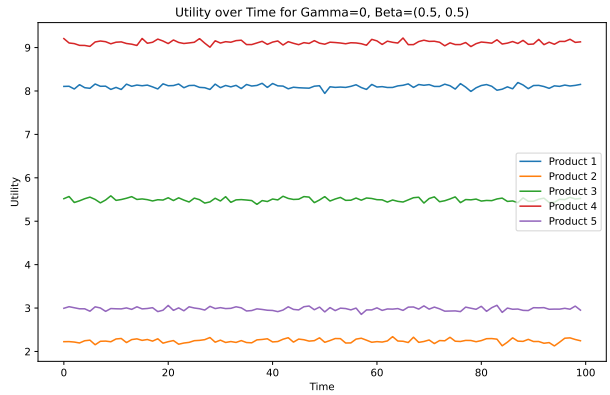

Figure 3: Utility graph for $\gamma = 0$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

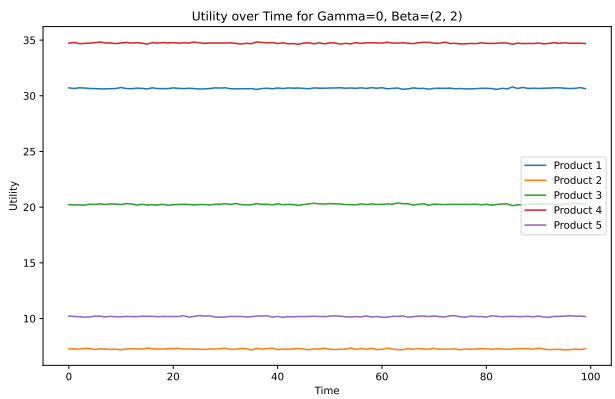

Figure 4: Utility graph for $\gamma = 0$, $\beta_1 = 2$, $\beta_2 = 2$

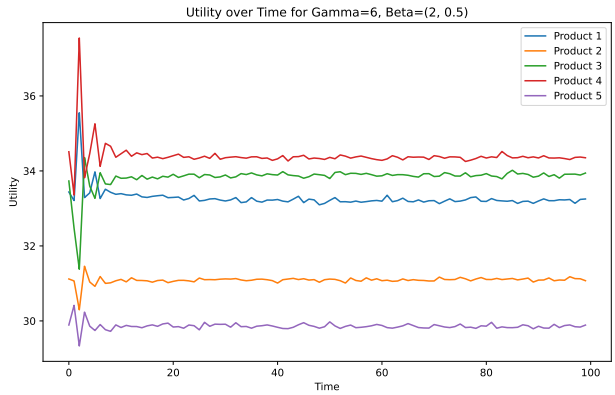

Figure 5: Utility graph for $\gamma = 6$, $\beta_1 = 2$, $\beta_2 = 0.5$

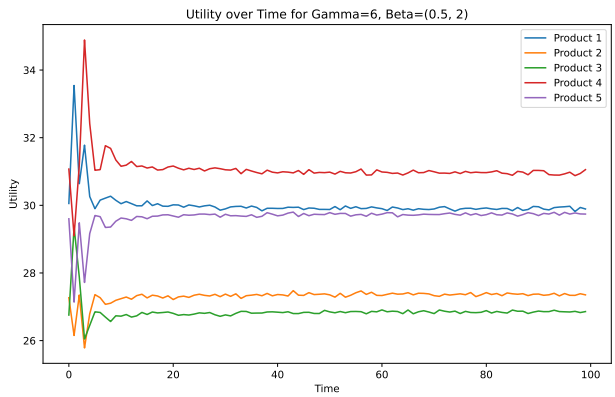

Figure 6: Utility graph for $\gamma = 6$, $\beta_1 = 0.5$, $\beta_2 = 2$

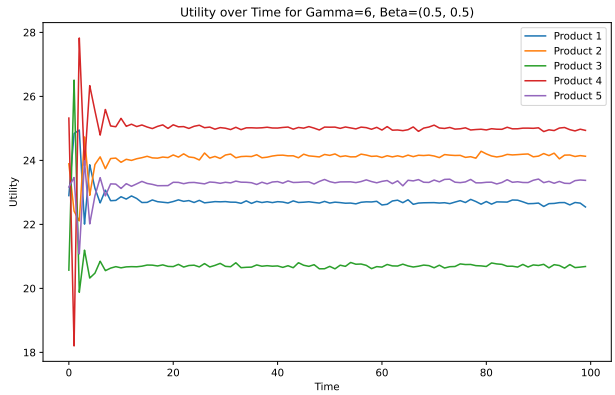

Figure 7: Utility graph for $\gamma = 6$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

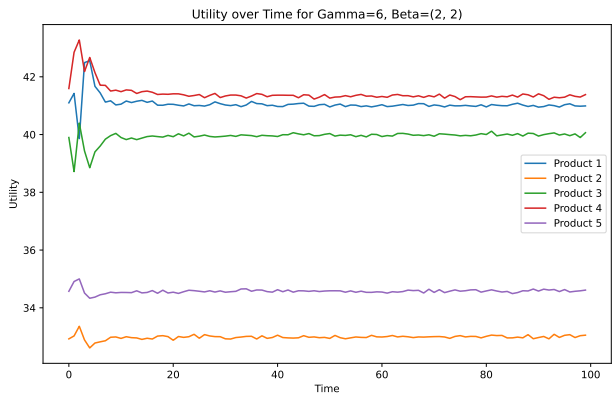

Figure 8: Utility graph for $\gamma = 6$, $\beta_1 = 2$, $\beta_2 = 2$

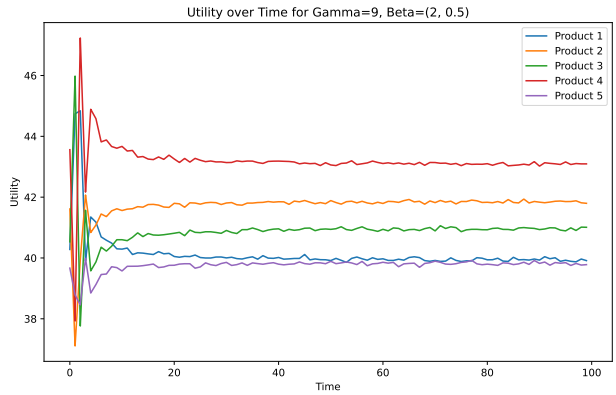

Figure 9: Utility graph for $\gamma = 9$, $\beta_1 = 2$, $\beta_2 = 0.5$

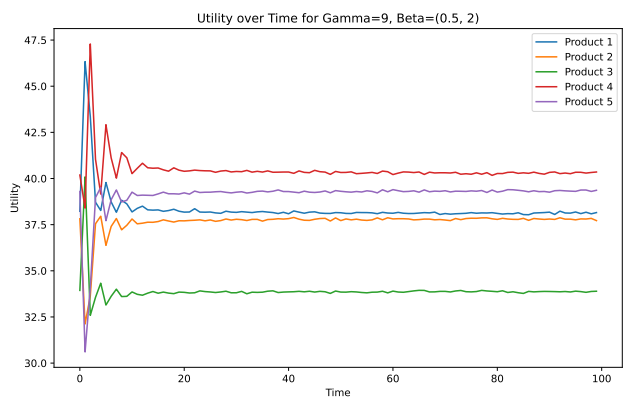

Figure 10: Utility graph for $\gamma = 9$, $\beta_1 = 0.5$, $\beta_2 = 2$

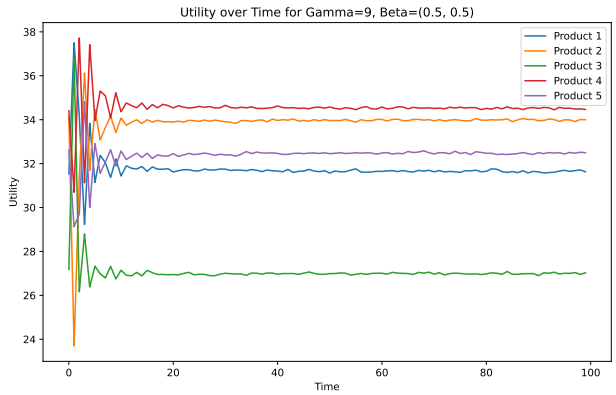

Figure 11: Utility graph for $\gamma = 9$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

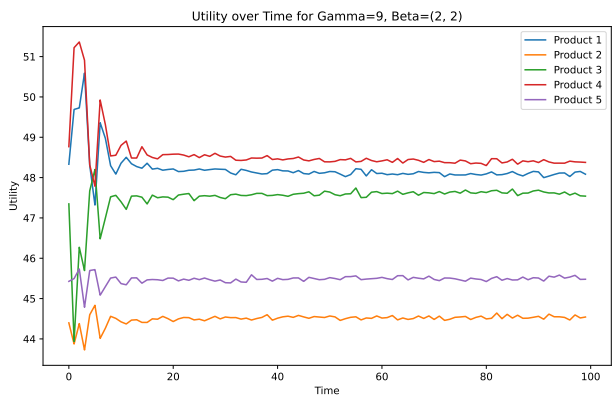

Figure 12: Utility graph for $\gamma = 9$, $\beta_1 = 2$, $\beta_2 = 2$

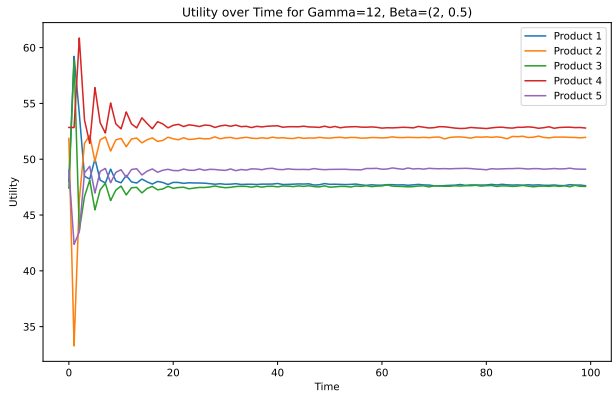

Figure 13: Utility graph for $\gamma = 12$, $\beta_1 = 2$, $\beta_2 = 0.5$

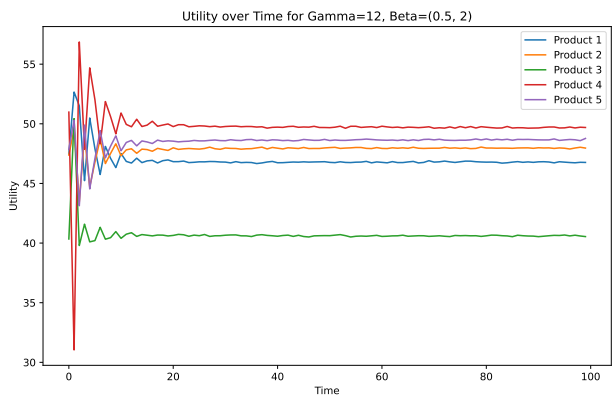

Figure 14: Utility graph for $\gamma = 12$, $\beta_1 = 0.5$, $\beta_2 = 2$

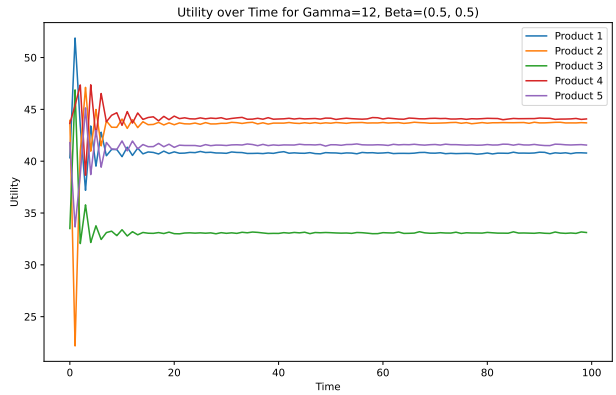

Figure 15: Utility graph for $\gamma = 12$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

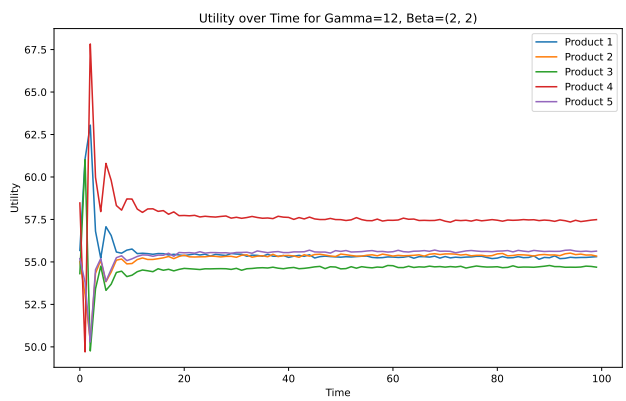

Figure 16: Utility graph for $\gamma = 12$, $\beta_1 = 2$, $\beta_2 = 2$

2 Regression Results on Log Difference in Shares

Dep. Variable:	y	**R-squared:**	0.990
Model:	OLS	**Adj. R-squared:**	0.990
Method:	Least Squares	**F-statistic:**	4.871e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	-889.94
No. Observations:	1000	**AIC:**	1786.
Df Residuals:	997	**BIC:**	1801.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0714	0.021	-3.474	0.001	-0.112	-0.031
x1	1.6305	0.006	295.524	0.000	1.620	1.641
x2	0.3398	0.005	73.380	0.000	0.331	0.349

Omnibus:	96.340	**Durbin-Watson:**	1.562
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	605.803
Skew:	0.121	**Prob(JB):**	2.83e-132
Kurtosis:	6.805	**Cond. No.**	6.20

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 2: Regression results for $y = \sum_j^2 \beta_j X_j + \varepsilon_j$ with $\beta_1 = 2$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.994
Model:	OLS	**Adj. R-squared:**	0.994
Method:	Least Squares	**F-statistic:**	7.851e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	-909.02
No. Observations:	1000	**AIC:**	1824.
Df Residuals:	997	**BIC:**	1839.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.2366	0.021	11.293	0.000	0.196	0.278
x1	0.3798	0.006	67.540	0.000	0.369	0.391
x2	1.7275	0.005	365.959	0.000	1.718	1.737

Omnibus:	280.928	**Durbin-Watson:**	1.126
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	2536.514
Skew:	1.018	**Prob(JB):**	0.00
Kurtosis:	10.532	**Cond. No.**	6.20

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 3: Regression results for $y = \sum_j^2 \beta_j X_j + \varepsilon_j$ with $\beta_1 = 0.5$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.997
Model:	OLS	**Adj. R-squared:**	0.997
Method:	Least Squares	**F-statistic:**	1.432e+05
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	967.58
No. Observations:	1000	**AIC:**	-1929.
Df Residuals:	997	**BIC:**	-1914.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.0155	0.003	4.836	0.000	0.009	0.022
x1	0.3570	0.001	414.642	0.000	0.355	0.359
x2	0.3669	0.001	507.628	0.000	0.365	0.368

Omnibus:	28.520	**Durbin-Watson:**	1.417
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	33.752
Skew:	0.345	**Prob(JB):**	4.69e-08
Kurtosis:	3.579	**Cond. No.**	6.20

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 4: Regression results for $y = \sum_j^2 \beta_j X_j + \varepsilon_j$ with $\beta_1 = 0.5$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.991
Model:	OLS	**Adj. R-squared:**	0.991
Method:	Least Squares	**F-statistic:**	5.330e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	-1094.8
No. Observations:	1000	**AIC:**	2196.
Df Residuals:	997	**BIC:**	2210.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0631	0.025	-2.499	0.013	-0.113	-0.014
x1	1.6780	0.007	247.783	0.000	1.665	1.691
x2	1.7747	0.006	312.208	0.000	1.764	1.786

Omnibus:	62.124	**Durbin-Watson:**	0.916
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	204.472
Skew:	0.215	**Prob(JB):**	3.98e-45
Kurtosis:	5.173	**Cond. No.**	6.20

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 5: Regression results for $y = \sum_j^2 \beta_j X_j + \varepsilon_j$ with $\beta_1 = 2$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.980
Model:	OLS	**Adj. R-squared:**	0.980
Method:	Least Squares	**F-statistic:**	1.600e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	-613.67
No. Observations:	1000	**AIC:**	1235.
Df Residuals:	996	**BIC:**	1255.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0034	0.016	-0.220	0.826	-0.034	0.027
x1	1.5522	0.007	218.803	0.000	1.538	1.566
x2	0.3879	0.004	106.073	0.000	0.381	0.395
x3	4.7520	0.026	180.096	0.000	4.700	4.804

Omnibus:	1426.909	**Durbin-Watson:**	1.838
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	587956.102
Skew:	-7.756	**Prob(JB):**	0.00
Kurtosis:	120.772	**Cond. No.**	10.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 6: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 6$, $\beta_1 = 2$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.981
Model:	OLS	**Adj. R-squared:**	0.981
Method:	Least Squares	**F-statistic:**	1.746e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	-841.16
No. Observations:	1000	**AIC:**	1690.
Df Residuals:	996	**BIC:**	1710.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0196	0.021	-0.946	0.344	-0.060	0.021
x1	0.3909	0.005	73.796	0.000	0.381	0.401
x2	1.5830	0.007	226.892	0.000	1.569	1.597
x3	4.7564	0.025	190.229	0.000	4.707	4.805

Omnibus:	351.901	**Durbin-Watson:**	1.455
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	85710.521
Skew:	0.187	**Prob(JB):**	0.00
Kurtosis:	48.353	**Cond. No.**	8.72

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 7: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 6$, $\beta_1 = 0.5$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.985
Model:	OLS	**Adj. R-squared:**	0.985
Method:	Least Squares	**F-statistic:**	2.143e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:47	**Log-Likelihood:**	-630.82
No. Observations:	1000	**AIC:**	1270.
Df Residuals:	996	**BIC:**	1289.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0133	0.016	-0.821	0.412	-0.045	0.018
x1	0.3911	0.004	90.525	0.000	0.383	0.400
x2	0.4101	0.004	114.023	0.000	0.403	0.417
x3	4.8198	0.020	237.330	0.000	4.780	4.860

Omnibus:	287.972	**Durbin-Watson:**	1.002
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	4162.495
Skew:	0.905	**Prob(JB):**	0.00
Kurtosis:	12.830	**Cond. No.**	8.09

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 8: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 6$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.988
Model:	OLS	**Adj. R-squared:**	0.988
Method:	Least Squares	**F-statistic:**	2.666e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-978.56
No. Observations:	1000	**AIC:**	1965.
Df Residuals:	996	**BIC:**	1985.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0692	0.023	-3.045	0.002	-0.114	-0.025
x1	1.6989	0.007	228.906	0.000	1.684	1.713
x2	1.7176	0.006	264.834	0.000	1.705	1.730
x3	5.2889	0.022	237.284	0.000	5.245	5.333

Omnibus:	758.462	**Durbin-Watson:**	1.105
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	68421.262
Skew:	2.752	**Prob(JB):**	0.00
Kurtosis:	43.147	**Cond. No.**	6.83

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 9: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 6$, $\beta_1 = 2$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.959
Model:	OLS	**Adj. R-squared:**	0.959
Method:	Least Squares	**F-statistic:**	7859.
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1340.9
No. Observations:	1000	**AIC:**	2690.
Df Residuals:	996	**BIC:**	2709.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.0028	0.033	0.085	0.932	-0.061	0.067
x1	1.6177	0.012	134.873	0.000	1.594	1.641
x2	0.4161	0.007	57.176	0.000	0.402	0.430
x3	7.4530	0.053	141.534	0.000	7.350	7.556

Omnibus:	930.166	**Durbin-Watson:**	1.670
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	504567.735
Skew:	3.282	**Prob(JB):**	0.00
Kurtosis:	112.848	**Cond. No.**	10.3

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 10: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 9$, $\beta_1 = 2$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.983
Model:	OLS	**Adj. R-squared:**	0.983
Method:	Least Squares	**F-statistic:**	1.913e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1191.0
No. Observations:	1000	**AIC:**	2390.
Df Residuals:	996	**BIC:**	2410.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0182	0.029	-0.629	0.530	-0.075	0.039
x1	0.3947	0.007	52.851	0.000	0.380	0.409
x2	1.6726	0.008	206.109	0.000	1.657	1.689
x3	7.5802	0.034	225.318	0.000	7.514	7.646

Omnibus:	268.813	**Durbin-Watson:**	1.543
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	19175.058
Skew:	0.052	**Prob(JB):**	0.00
Kurtosis:	24.452	**Cond. No.**	8.10

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 11: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 9$, $\beta_1 = 0.5$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.985
Model:	OLS	**Adj. R-squared:**	0.985
Method:	Least Squares	**F-statistic:**	2.194e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1045.5
No. Observations:	1000	**AIC:**	2099.
Df Residuals:	996	**BIC:**	2119.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0388	0.024	-1.587	0.113	-0.087	0.009
x1	0.3997	0.006	61.719	0.000	0.387	0.412
x2	0.4323	0.005	79.867	0.000	0.422	0.443
x3	7.6032	0.031	243.037	0.000	7.542	7.665

Omnibus:	261.756	**Durbin-Watson:**	1.079
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	8942.167
Skew:	0.477	**Prob(JB):**	0.00
Kurtosis:	17.619	**Cond. No.**	8.20

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 12: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 9$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.978
Model:	OLS	**Adj. R-squared:**	0.978
Method:	Least Squares	**F-statistic:**	1.487e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1489.7
No. Observations:	1000	**AIC:**	2987.
Df Residuals:	996	**BIC:**	3007.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.1438	0.038	-3.792	0.000	-0.218	-0.069
x1	1.7455	0.012	150.102	0.000	1.723	1.768
x2	1.7827	0.010	179.060	0.000	1.763	1.802
x3	8.0777	0.043	188.821	0.000	7.994	8.162

Omnibus:	530.899	**Durbin-Watson:**	0.997
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	61383.864
Skew:	-1.440	**Prob(JB):**	0.00
Kurtosis:	41.274	**Cond. No.**	7.41

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 13: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 9$, $\beta_1 = 2$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.980
Model:	OLS	**Adj. R-squared:**	0.980
Method:	Least Squares	**F-statistic:**	1.634e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1318.2
No. Observations:	1000	**AIC:**	2644.
Df Residuals:	996	**BIC:**	2664.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0630	0.032	-1.970	0.049	-0.126	-0.000
x1	1.6837	0.010	162.819	0.000	1.663	1.704
x2	0.4516	0.007	63.396	0.000	0.438	0.466
x3	10.4125	0.050	207.486	0.000	10.314	10.511

Omnibus:	273.362	**Durbin-Watson:**	1.335
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	20386.286
Skew:	-0.101	**Prob(JB):**	0.00
Kurtosis:	25.119	**Cond. No.**	9.97

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 14: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 12$, $\beta_1 = 2$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.983
Model:	OLS	**Adj. R-squared:**	0.983
Method:	Least Squares	**F-statistic:**	1.975e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1457.6
No. Observations:	1000	**AIC:**	2923.
Df Residuals:	996	**BIC:**	2943.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0033	0.038	-0.089	0.929	-0.077	0.071
x1	0.4034	0.010	41.408	0.000	0.384	0.422
x2	1.7370	0.010	181.590	0.000	1.718	1.756
x3	10.4404	0.045	232.791	0.000	10.352	10.528

Omnibus:	454.432	**Durbin-Watson:**	1.777
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	67666.243
Skew:	0.998	**Prob(JB):**	0.00
Kurtosis:	43.249	**Cond. No.**	8.14

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 15: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 12$, $\beta_1 = 0.5$, $\beta_2 = 2$

Dep. Variable:	y	**R-squared:**	0.978
Model:	OLS	**Adj. R-squared:**	0.978
Method:	Least Squares	**F-statistic:**	1.478e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1552.5
No. Observations:	1000	**AIC:**	3113.
Df Residuals:	996	**BIC:**	3133.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.0751	0.041	-1.855	0.064	-0.155	0.004
x1	0.4119	0.011	38.419	0.000	0.391	0.433
x2	0.4550	0.009	50.503	0.000	0.437	0.473
x3	10.3888	0.052	200.191	0.000	10.287	10.491

Omnibus:	771.524	**Durbin-Watson:**	1.515
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	218105.891
Skew:	-2.476	**Prob(JB):**	0.00
Kurtosis:	75.180	**Cond. No.**	8.17

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 16: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 12$, $\beta_1 = 0.5$, $\beta_2 = 0.5$

Dep. Variable:	y	**R-squared:**	0.974
Model:	OLS	**Adj. R-squared:**	0.974
Method:	Least Squares	**F-statistic:**	1.227e+04
Date:	Thu, 07 May 2026	**Prob (F-statistic):**	0.00
Time:	16:32:48	**Log-Likelihood:**	-1774.0
No. Observations:	1000	**AIC:**	3556.
Df Residuals:	996	**BIC:**	3576.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-0.1393	0.051	-2.756	0.006	-0.238	-0.040
x1	1.7299	0.015	117.268	0.000	1.701	1.759
x2	1.7970	0.012	144.273	0.000	1.773	1.821
x3	10.7567	0.061	176.425	0.000	10.637	10.876

Omnibus:	1029.382	**Durbin-Watson:**	1.025
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	215820.737
Skew:	-4.315	**Prob(JB):**	0.00
Kurtosis:	74.451	**Cond. No.**	7.83

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 17: $y = \sum_j^2 \beta_j X_j + \gamma \log(1 + \Xi^2)$ with $\gamma = 12$, $\beta_1 = 2$, $\beta_2 = 2$